SPOTIFLOW PLATFORM

Databricks Delta Lake Medallion Architecture & FinOps Defense

PHASE 3 FINAL

1.29 T	26 M	65.4%	44.86
TOTAL STREAMS	ACTIVE TRACKS	PIPELINE SUCCESS	AVG PEAK POSITION

1. EXECUTIVE SUMMARY & ARCHITECTURE

The **Spotiflow** analytics platform is an end-to-end data engineering pipeline designed to ingest, cleanse, aggregate, and visualize high-volume streaming charts across global regions. Built on Azure Databricks using the **Delta Lake Medallion Pattern**, the platform guarantees ACID compliance, idempotent incremental loading, and sub-second analytical querying in Power BI.



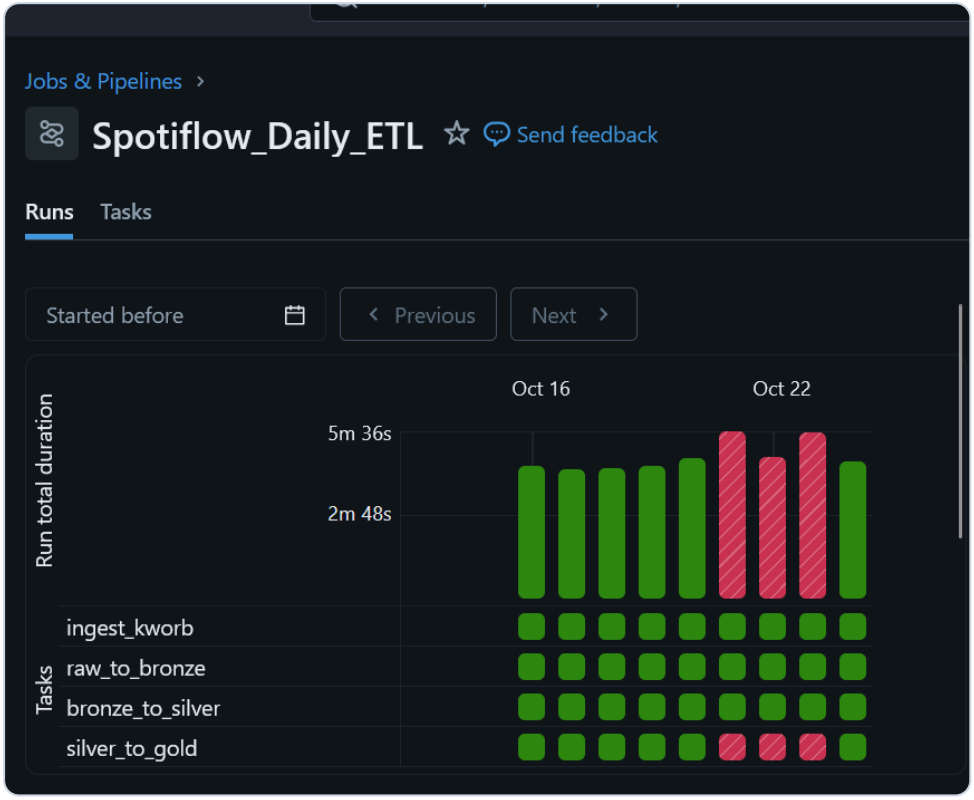
2. BUSINESS INTELLIGENCE DASHBOARD OVERVIEW

The Gold layer is connected directly to a custom dark-themed Power BI dashboard that provides real-time visibility into global streaming trends and pipeline operational health.



3. PIPELINE AUTOMATION & INCIDENT DEFENSE

Pipeline orchestration is automated via Databricks Workflows (`Spotiflow_Daily_ETL`), running four sequential tasks: `ingest_kworb` → `raw_to_bronze` → `bronze_to_silver` → `silver_to_gold`. Audit logging is tracked in real-time within the `Pipeline logs` table.



Engineering Incident & Historical Backfill Analysis

During pipeline testing, a 3-day execution failure occurred in `silver_to_gold` due to an upstream schema modification in the Bronze layer (column renaming). The pipeline logging captured the failure state (`FAILED`, 0 rows inserted), preventing invalid propagation. The Bronze ingestion code was patched with explicit casting, and an automated **historical backfill script** was executed across the 3 affected processing dates. Thanks to Delta Lake `MERGE` idempotency, all missing dates were re-processed and caught up cleanly without duplicate row creation.

4. GOLD LAYER STAR SCHEMA MODELING

The Gold layer implements a normalized **Star Schema** to decouple analytical reporting from upstream ingestion structures.

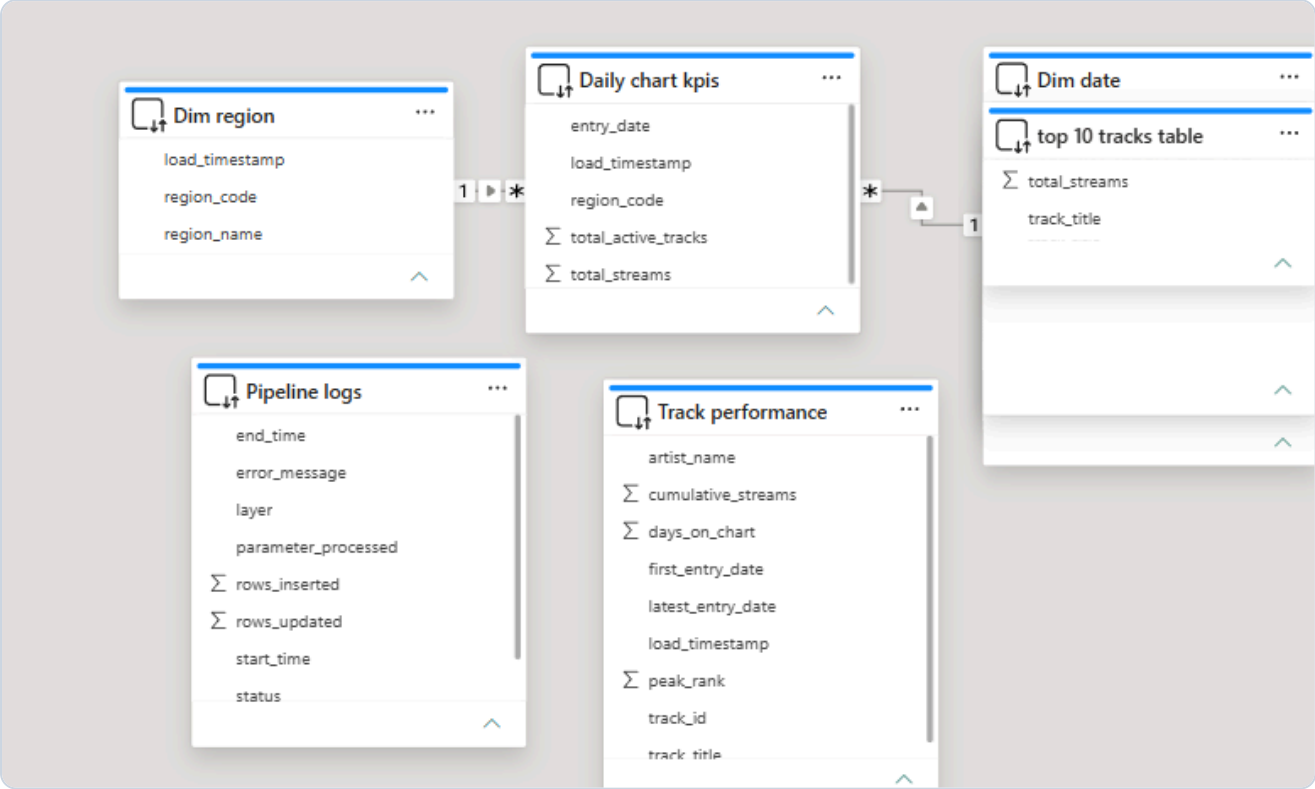
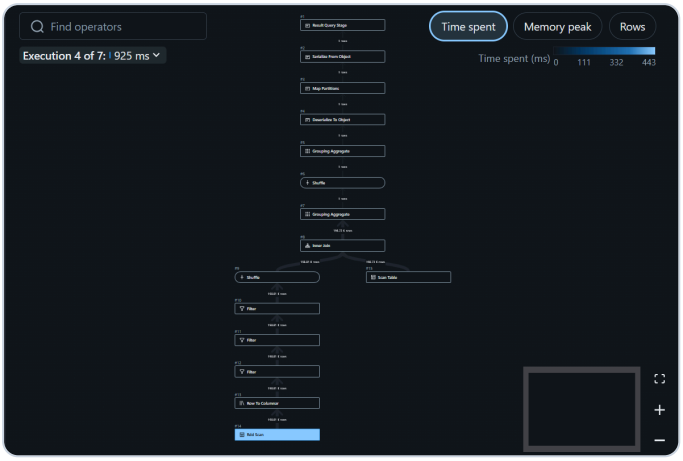


Table Name	Type	Keys	Description
Daily chart kpis	Fact Table	`entry_date`, `region_code` (FK)	Aggregated daily chart metrics including total active tracks and total streams.
Track performance	Fact / Aggregate	`track_id` (PK)	Cumulative streaming performance, peak rank, and days on chart per track.
Dim region	Dimension	`region_code` (PK)	Conformed region lookup table mapping codes to full region names.
Dim date	Dimension	`date_key` (PK)	Standard calendar dimension supporting multi-year date slicers.
Pipeline logs	Audit Console	`start_time`	Operational metadata tracking layer, status, duration, and row counts.

5. SPARK UI & DAG EXECUTION ANALYSIS

Analysis of the Apache Spark execution DAG during the Gold layer aggregation query (`merge_track_re...`, 48 tasks, 12.99s runtime) reveals optimal distributed execution patterns:



- **Predicate Pushdown (Operators #10, #11, #12):** Spark pushes filter criteria down to the file scan layer (`Rdd Scan` / `Scan Table`), pruning non-target date partitions before reading data into memory.
- **Shuffle Exchange (Operators #6, #9):** Data repartitioning occurs prior to the `Inner Join (#8)` (combining 198.81K rows with 198.73K rows) and final `Grouping Aggregate (#5)` across worker nodes.
- **Two-Stage Aggregation:** Local partial aggregation (`Grouping Aggregate #7`) executes on individual nodes before network shuffle (#6), drastically reducing data transferred across the cluster.

6. FINOPS EVALUATION & ANNUAL COST PROJECTIONS

Compute consumption was evaluated based on single-node cluster runs for `Spotiflow_Daily_ETL` (average duration: 2 min 48 sec to 5 min 36 sec per run).

Execution Cadence	Cluster Specification	Avg Duration	Estimated Cost
Daily Batch Run (Current)	Standard_DS3_v2 (4 cores, 14GB)	~ 3.5 minutes	~$0.064 USD / run
Hourly Enterprise Batch	Automated Job Cluster	8,760 runs / yr	**~$560.64 USD / year**

7. FINOPS OPTIMIZATION RECOMMENDATIONS

Enterprise Cost Reduction Levers

1. **Job Clusters vs All-Purpose:** Executing workflows on automated Job Clusters rather than interactive compute reduces DBU costs by up to 50%.
2. **Scheduled Table Maintenance:** Execute `OPTIMIZE ZORDER BY` and `VACUUM` on a weekly schedule off-peak rather than after every micro-batch run.
3. **Serverless SQL for Power BI:** Connect Power BI DirectQuery to Databricks Serverless SQL Warehouses to leverage immediate auto-suspend when queries idle.